

Verified STEAD identities

2234 records / 1477 sources
Numerical waveform matching
E, N, Z; 100 Hz; 60 s



Source-disjoint partition

Train 1034 / validation 222 / test 221 sources
1544 / 360 / 330 records
One frozen split; zero source overlap



Temporal meta-learner

12 picker-score + 3 waveform channels
Meta seeds 42, 43, 44 within each base
Outer validation reused across final stages



Fixed arrival decoder

P maximum; S search P + 0.1–30 s
Detected and uncertain accepted
Detected-only sensitivity retained



Base models and OOF

STA/LTA, AIC, CNN, TCN
Five source folds with inner validation
Base seeds 42, 43, 44

Source-held-out evaluation

F1 at 50, 100, 200 ms; accepted-pick MAE
Full3C / Z-only; ablations; PhaseNet-style
Within-base means; source-cluster intervals